

PRACTICAL 7

Linux Process Address Space and Virtual Memory Analysis

TASK 1

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Screenshot 1 – Successful compilation of the Task 1 program.

```
root@HP:~# nano task1.c
root@HP:~# nano task1.c
root@HP:~# gcc task1.c -o task1
root@HP:~# |
```

Screenshot 2 – Task 1 output showing the addresses of code, global, static, heap, and stack variables.

```
root@HP:~# ./task1
Linux Process Memory Address Space
-----
Code address      : 0x583c6ff751c9
Global variable   : 0x583c6ff78010
Static variable   : 0x583c6ff78014
Heap variable     : 0x583ca9ac32a0
Stack variable    : 0x7ffcd6bf164c
Code address      : 0x583c6ff751c9
root@HP:~# |
```

TASK 2

Make use of the `/proc/<PID>/maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Screenshot 3 – Successful compilation of the Task 2 program.

```

root@HP:~# gcc task2.c -o task2
root@HP:~# nano task2.c
root@HP:~# gcc task2.c -o task2
root@HP:~# |

```

Screenshot 4 – Task 2 running process showing PID and memory addresses.

```

root@HP:~# gcc task2.c -o task2
root@HP:~# ./task2
Process ID (PID): 906
Code address      : 0x622f154471c9
Global address    : 0x622f1544a010
Static address    : 0x622f1544a014
Heap address      : 0x70f09d62e010
Stack address     : 0x7fffd5ec0caec

Process is running...
|

```

Screenshot 5 – /proc/906/maps displaying the virtual memory mappings of the process.

```

root@HP:~# cat /proc/906/maps
622f15446000-622f15447000 r--p 00000000 08:30 55413 /root/task2
622f15447000-622f15448000 r-xp 00001000 08:30 55413 /root/task2
622f15448000-622f15449000 r--p 00002000 08:30 55413 /root/task2
622f15449000-622f1544a000 r--p 00002000 08:30 55413 /root/task2
622f1544a000-622f1544b000 rw-p 00003000 08:30 55413 /root/task2
622f1e58d000-622f1e5ae000 rw-p 00000000 00:00 0 [heap]
70f09d400000-70f09d428000 r--p 00000000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d428000-70f09d5b0000 r-xp 00028000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d5b0000-70f09d5ff000 r--p 001b0000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d5ff000-70f09d603000 r--p 001fe000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d603000-70f09d605000 rw-p 00202000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d605000-70f09d612000 rw-p 00000000 00:00 0
70f09d62e000-70f09d732000 rw-p 00000000 00:00 0
70f09d737000-70f09d739000 rw-p 00000000 00:00 0
70f09d739000-70f09d73d000 r--p 00000000 00:00 0 [vvar]
70f09d73d000-70f09d73f000 r--p 00000000 00:00 0 [vvar_vclock]
70f09d73f000-70f09d741000 r-xp 00000000 00:00 0 [vdso]
70f09d741000-70f09d742000 r--p 00000000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86_64.so.2
70f09d742000-70f09d76d000 r-xp 00001000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86_64.so.2
70f09d76d000-70f09d777000 r--p 0002c000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86_64.so.2
70f09d777000-70f09d779000 r--p 00036000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86_64.so.2
70f09d779000-70f09d77b000 rw-p 00038000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86_64.so.2
7fffd5ebee000-7fffd5ec0f000 rw-p 00000000 00:00 0 [stack]
root@HP:~#

```

Screenshot 6 – Filtered mappings showing the executable, heap, and stack regions.

```

root@HP:~# cat /proc/906/maps
622f15446000-622f15447000 r--p 00000000 08:30 55413 /root/task2
622f15447000-622f15448000 r-xp 00001000 08:30 55413 /root/task2
622f15448000-622f15449000 r--p 00002000 08:30 55413 /root/task2
622f15449000-622f1544a000 r--p 00002000 08:30 55413 /root/task2
622f1544a000-622f1544b000 rw-p 00003000 08:30 55413 /root/task2
622f1e58d000-622f1e5ae000 rw-p 00000000 00:00 0 [heap]
70f09d400000-70f09d428000 r--p 00000000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d428000-70f09d5b0000 r-xp 00028000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d5b0000-70f09d5ff000 r--p 001b0000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d5ff000-70f09d603000 r--p 001fe000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d603000-70f09d605000 rw-p 00202000 08:30 18083 /usr/lib/x86_64-linux-gnu/libc.so.6
70f09d605000-70f09d612000 rw-p 00000000 00:00 0
70f09d62e000-70f09d732000 rw-p 00000000 00:00 0
70f09d732000-70f09d739000 rw-p 00000000 00:00 0
70f09d739000-70f09d73d000 r--p 00000000 00:00 0 [vvar]
70f09d73d000-70f09d73f000 r--p 00000000 00:00 0 [vvar_vclock]
70f09d73f000-70f09d741000 r-xp 00000000 00:00 0 [vdso]
70f09d741000-70f09d742000 r--p 00000000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70f09d742000-70f09d76d000 r-xp 00001000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70f09d76d000-70f09d777000 r--p 0002c000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70f09d777000-70f09d779000 r--p 00036000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70f09d779000-70f09d77b000 rw-p 00038000 08:30 18080 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffd5eb000-7ffd5ec0f000 rw-p 00000000 00:00 0 [stack]
root@HP:~#

```

Screenshot 7 – Process memory information from /proc/906/status.

```

root@HP:~# cat /proc/906/maps | grep -E '\[heap\]|\[stack\]|task2'
622f15446000-622f15447000 r--p 00000000 08:30 55413 /root/task2
622f15447000-622f15448000 r-xp 00001000 08:30 55413 /root/task2
622f15448000-622f15449000 r--p 00002000 08:30 55413 /root/task2
622f15449000-622f1544a000 r--p 00002000 08:30 55413 /root/task2
622f1544a000-622f1544b000 rw-p 00003000 08:30 55413 /root/task2
622f1e58d000-622f1e5ae000 rw-p 00000000 00:00 0 [heap]
7ffd5eb000-7ffd5ec0f000 rw-p 00000000 00:00 0 [stack]
root@HP:~#

```

Screenshot 8 – pmap displaying the memory regions and total memory of the process.

```

root@HP:~# cat /proc/906/status | grep -E 'VmSize|VmRSS|VmData|VmStk|VmExe'
VmSize:      3716 kB
VmRSS:       1676 kB
VmData:      1252 kB
VmStk:       132 kB
VmExe:       4 kB
root@HP:~#

```

Screenshot 9 – ps displaying the virtual and resident memory usage of the process.

```

root@HP:~# pmap 906
906:      ./task2
0000622f15446000      4K r---- task2
0000622f15447000      4K r-x-- task2
0000622f15448000      4K r---- task2
0000622f15449000      4K r---- task2
0000622f1544a000      4K rw--- task2
0000622f1e58d000     132K rw--- [ anon ]
000070f09d400000     160K r---- libc.so.6
000070f09d428000    1568K r-x-- libc.so.6
000070f09d5b0000     316K r---- libc.so.6
000070f09d5ff000      16K r---- libc.so.6
000070f09d603000       8K rw--- libc.so.6
000070f09d605000      52K rw--- [ anon ]
000070f09d62e000    1040K rw--- [ anon ]
000070f09d737000       8K rw--- [ anon ]
000070f09d739000      16K r---- [ anon ]
000070f09d73d000       8K r---- [ anon ]
000070f09d73f000       8K r-x-- [ anon ]
000070f09d741000       4K r---- ld-linux-x86-64.so.2
000070f09d742000     172K r-x-- ld-linux-x86-64.so.2
000070f09d76d000      40K r---- ld-linux-x86-64.so.2
000070f09d777000       8K r---- ld-linux-x86-64.so.2
000070f09d779000       8K rw--- ld-linux-x86-64.so.2
00007fffd5ebee000    132K rw--- [ stack ]
total                 3716K
root@HP:~#

```

Screenshot 10 – top displaying the running process and its memory usage.

```

root@HP:~# ps -p 906 -o pid,comm,vsz,rss
  PID COMMAND      VSZ   RSS
   906 task2        3716  1676
root@HP:~#

```

FINAL OBSERVATION

The practical demonstrated the Linux process address space and virtual memory organization. The addresses of code, global, static, heap, and stack variables were observed, and the virtual memory mappings of a running process were studied using /proc/906/maps and memory analysis tools.